

Grace Elisabeth Coppinger Sinclair

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RESEARCH INTERESRTS

I am interested in understanding how climate change and urbanization impact host and pathogen relationships, distribution, and evolution. Currently, I utilize mathematical and molecular approaches to answer these questions in a common weedy plant (*Plantago rugelli*) and their fungal pathogens (*Golovinomyces sordidus*). I am completing my Ph.D. in the lab of Rachel Penczykowski and am on track to graduate in 2030.

RESEARCH EXPERIENCE

Ph.D. Student Researcher — *Washington University in Saint Louis, Ecology and Evolutionary Biology*

- 2024-Present | Principal Investigator: Rachel Penczykowski
- “Understanding the temporal and spatial impacts of climate change and urbanization on a wild plant-pathogen system.”

Honors Undergraduate Researcher — *University of South Alabama, Department of Biology*

- 2021-2024 | Principal Investigator: Jason Strickland
- “*Thamnophis sirtalis* and their toxic relationship: Testing for intraspecific venom variation in Common Garter Snakes.”

Research Assistant — *University of South Alabama, Department of Biology*

- 2022-2023 | Principal Investigator: Jason Strickland in collaboration with Megan Hermance
- “A Ticking Time Bomb: Testing how viral infection modifies salivary gland gene expression in an invasive tick during feeding.”

Research Assistant — *University of South Alabama, Department of Civil, Costal, and Environmental Engineering*

- 2022-2023 | Principal Investigator: Kevin White
- “Decentralized Wastewater Technology and Management Applications in the Rural Black Belt Alabama.”

EDUCATION

Washington University in Saint Louis — *Ph.D. Ecology and Evolutionary Biology*

August 2024 - Present | Saint Louis, Missouri

University of South Alabama — *B.S. Biology*

August 2020 - May 2024 | Mobile, Alabama

FELLOWSHIPS AND GRANTS

Danforth Fellowship — *Washington University in Saint Louis*

- Awarded in 2025 | Principal Investigator: Rachel Penczykowski
- “Forecasting and validating disease hotspots in a natural plant–pathogen system.”

National Science Foundation Research Experience for Undergraduates — *Cornell University, Boyce Thompson Institute*

- Awarded in 2023 | Principal Investigator: Zhangjun Fei
- “Identification of structural variants (SVs) in wild and cultivated watermelon.”

Summer Undergraduate Research Experience — *University of South Alabama*

- Awarded in 2022 | Principal Investigator: Jason Strickland
- “Reptile and amphibian diversity on the campus of the University of South Alabama.”

PUBLICATIONS

Coppinger, Grace E., Aaron J. Stewart, Joel A. Borden, and Jason L. Strickland. "Thamnophis sirtalis and their toxic relationship: Testing for intraspecific venom variation in Common Garter Snakes." *Toxicon* 253 (2025): 108185.

Canterbury, D.N, **Grace E. Coppinger**, Sammie Morris, Jason L. Strickland. "Hyla gratiosa (Barking Treefrog) axanthism." *Herpetological Review* 54(2), 266–267 (2023).

PRESENTATIONS

Coppinger, G.E., A.J. Stewart, and J.L. Strickland. *Thamnophis sirtalis* and their toxic relationship: Testing for intraspecific venom variation in Common Garter Snakes. (2025) Ecology and Evolutionary Biology Seminar at Washington University in Saint Louis, Saint Louis, Missouri.

Coppinger, G.E., H. Sun, and Z. Fei. Identification of structural variants between wild and cultivated watermelons through whole genome comparison. (2023) BTI Kohut Symposium, Ithaca, NY.

Stewart, A.J., **G.E Coppinger**, J.A Borden and J.L. Strickland. Diet diversity drives phenotypic variability in the venom of rear-fanged snakes. (2023) 70th annual meeting of the Southwestern Association for Naturalist, San Antonio, Texas. 2nd Place.

Coppinger, G.E., A.J. Stewart, and J.L. Strickland. Understanding Venom Transcriptomics of the Rear-fanged Snake *Thamnophis sirtalis*. (2022) Undergraduate Research Symposium at The University of South Alabama, Mobile, Alabama.

Coppinger, G.E., J.L. Strickland, and J.A. Borden. Reptile and Amphibian Diversity on the University of South Alabama’s Campus. (2021) Meeting of the Alabama Partners in Amphibian and Reptile Conservation (ALAPARC), Andalusia, Alabama.

Coppinger, G.E., J.L. Strickland, and J.A. Borden. Reptile and Amphibian Diversity on the University of South Alabama’s Campus. (2021) Meeting of the Southeast Partners in Amphibian and Reptile Conservation (SEPARC), Hosted Virtually.

Coppinger, G.E., J.L. Strickland, and J.A. Borden. Reptile and Amphibian Diversity on the University of South Alabama's Campus. (2021) Undergraduate Research Symposium at The University of South Alabama, Mobile, Alabama.

TEACHING EXPERIENCE

Teaching Assistant — "Disease Ecology" BIOL.4195.01 Washington University in St. Louis | Spring 2026

- Instructor: Rachel Penczykowski
- Responsibilities: Mentor undergraduate students in mathematical model design, data interpretation, and scientific communication. I graded quizzes, assignments, and lead a weekly discussion section that reviewed recent literature. Finally, I lead a lecture on the topic of "Community ecology of disease: Diversity-disease patterns".

Teaching Assistant — "Evolution" BIOL.3501.01 Washington University in St. Louis | Fall 2025

- Instructor: Ken Olsen
- Responsibilities: Assist undergraduate students in data interpretation, and scientific communication, and understanding of Evolutionary biology. I graded essays, exams, and lead a weekly discussion section that reviewed recent literature.

Learning Assistant — "Advanced Bioinformatics" BLY 490 University of South Alabama | Spring 2024

- Instructor: Jason Strickland
- Responsibilities: Assist undergraduate students in generation and understanding bioinformatic code.

MENTOR EXPERIENCE

Graduate student mentor for undergraduates (2025-present)

- So far in my graduate student career I have mentored 2 undergraduate students to conduct and complete their thesis research. My mentorship included assisting students with laboratory work, data analysis, data visualization, R training, mathematical model construction, writing manuscripts, and scientific presentations.
 - Elizabeth Grieve (2025-present): "Temperature Dependence in the Sexual Lifecycle of a Fungal Plant Pathogen Along a Latitudinal Gradient."
 - Auriel Prepejchal (2025-2026): "Mathematical modeling of temperature-sensitive *Golovinomyces sordidus* across wild *Plantago rugelii* patches over an urban heat gradient."

Tyson Mentor for the Plant disease team (2025)

- I mentored 3 undergraduate students and 2 high school students for a Tyson field summer. My mentorship included preparing students for field work, data analysis, and design of their independent research projects and presentations.

SSSTING Lab peer mentor (2021-2024)

- I mentored 7 undergraduates during my time in the SSSTING Lab. My mentorship included preparing students for literature review, RNA extraction, cDNA library prep, bioinformatic pipelines, and tutoring them for classes.

Go Explore Math and Science Outreach (2023)

- Local middle and elementary students visited the University of South Alabama, and I led a section on the identification and importance of local reptiles and amphibians.

PROFESSIONAL SERVICE AND LEADERSHIPS

EEB Student Representative (2025-Present)

- Serve as a liaison between the EEB faculty and graduate students for Washington University in Saint Louis.
- Plan and facilitate EEB graduate student activities

Member of the EEB seminar committee (2026-Present)

- Facilitate and plan weekly seminars accessible by Washington University faculty, students, and the general public.
- Generate and distribute weekly physical and digital flyers and promote the seminar.

Honors College Representative (2022-2024)

- Serve as a liaison between the Honors College and prospective students and University faculty.

SSSTING Lab Member (2021-2024)

- Perform research, outreach to local communities, and manage social media accounts

Biology Student Association Member (2021-2024)

- Attend meetings and outreach opportunities

Marine Science Student Association Member (2021-2023)

- Attend meetings and outreach opportunities

First Year Council Member (2020-2021)

- Awarded member of the year. Improve freshman campus experiences, work with local safety committees and police departments, create "Awareness Wednesday" informational social media outreach

Honors College Member (2020-2024)

- Propose and defend an honors Thesis, complete honors specific coursework, maintain 3.5 GPA, participate in local community outreach

ACADEMIC SCHOLARSHIP AND AWARDS

Jaguar Senior Medallion Award - Undergraduate research award

- The medallion recognizes graduating seniors who have demonstrated excellence through achievement in scholarship and engagement in activities at the University of South Alabama. First recipient of the undergraduate research award.

USA Honors College - Undergraduate Research Award

- Recognized graduating students in the honors college who demonstrated excellence in research.

Rawls Outstanding Senior Award

- Awarded to the most outstanding senior biology major in the Biology Department at the University of South Alabama.

Whiddon Honors Scholarship

- Full tuition funding provided to Honors College students displaying excellence

USA Honors College Exploration and Experiential Learning Scholarship

- Scholarship awarded to students apart of the Honors College who are dedicated to becoming a

well-rounded student through service.

Honors Summer Scholar Fellowship

- Scholarship awarded to students apart of the Honors College who are dedicated to furthering their academic achievement.

Ron Kiene Research Scholarship

- Scholarship awarded to students who are dedicated to protecting our environment through research.

Angelia and Steven Stokes Undergraduate Scholarship in Marine Sciences

- Scholarship awarded to Marine Science students who excel in research, coursework, and involvement on campus.

Mike deGruy Memorial Scholarship

- Scholarship awarded to Marine Science students who excel in research, coursework, and involvement on campus.

Brian Axsmith Memorial Scholarship

- Scholarship awarded by the Biology Department to students that are excelling in their classwork and research with interests in botany, ecology, environmental science, or evolution.

Neely Memorial Scholarship

- Scholarship awarded to students at the University of South Alabama who excel in research and academics.

Fresh Admission Scholar AB Tier 4

- Merit based scholarship

FIELD, LABRATORY, AND COMPUTATIONAL SKILLS

Computational and Bioinformatics

- R (data analysis, visualization, theoretical statistical modeling)
- Population genomic and bioinformatic analyses
- High-performance computing / command-line workflows
- Bioinformatics and genomic data analysis

Molecular and Genomic

- DNA and RNA extraction
- PCR and molecular biology techniques
- DNA/RNA quantification and quality assessment (Qubit, NanoDrop, TapeStation)
- RNA-seq and transcriptomic workflows
- Short- and long-read sequencing
- Genome assembly and comparative genomics
- Genomic/transcriptomic data analysis

Field

- Plant disease surveys
- Plant phenology and ecological monitoring
- Herpetological surveys

- Drift-fence and funnel-trap sampling
- Specimen collection and identification

REFERENCES

Rachel Penczykowski, Ph.D. | rpenczykowski@wustl.edu

- Associate Professor
- Department of Biology Washinton University in Saint Louis
- One Brookings Drive St. Louis, MO 63130

Kim Medley, Ph.D. | kim.medley@wustl.edu

- Director, Tyson Research Center
- Department of Biology Washinton University in Saint Louis
- 6750 Tyson Valley Road, Eureka, MO 63021

Jason Strickland, Ph.D | jasonstrickland@southalabama.edu

- Assistant Professor
- Department of Biology University of South Alabama
- 5871 USA Dr. N Mobile, AL 36688